# Monitoring Deforestation and Land-Use Change in the Amazon Using Deep Learning on Satellite Imagery

Data Science II — Project Proposal | Rank: 1st Choice (Recommended)

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## Introduction

We propose to build a deep learning system that automatically tags satellite image tiles of the Amazon basin with the land-cover, land-use, and atmospheric conditions visible in them — including primary rainforest, selective logging, slash-and-burn agriculture, artisanal and conventional mining, roads, cultivation, and habitation. Because a single image tile can show several of these conditions at once (for example, "clear" weather over a region with both "primary" forest and "agriculture"), this is a **multi-label** image classification problem rather than a simple single-class task.

Our planned approach includes:

- **Data preprocessing:** Normalize the 4-band (Red, Green, Blue, Near-Infrared) GeoTIFF/JPEG image chips, and engineer an additional Normalized Difference Vegetation Index (NDVI) channel from the Red and NIR bands to make vegetation health and canopy loss more visible to the model. We will also examine the frequency of each of the 17 possible labels to identify severe class imbalance (e.g., "blow_down" and "artisanal_mine" appear in under 1% of tiles) and apply targeted oversampling and augmentation (rotations and flips, which are natural for top-down satellite imagery) for rare classes.
- **Model training:** Build a convolutional neural network for multi-label classification with a sigmoid output layer and binary cross-entropy loss per label. We will compare a CNN trained from scratch against transfer learning from an ImageNet-pretrained backbone (e.g., ResNet or EfficientNet) adapted to accept 4+ input channels, and will tune hyperparameters such as learning rate, thresholding per label, and class weighting.
- **Evaluation:** Measure performance with per-label and macro-averaged metrics, analyze which labels are hardest to detect (typically the rare, high-impact classes such as mining and logging), and generate confusion analysis and visualizations of model attention (Grad-CAM) over misclassified tiles.

**Related work:** This problem is based on the well-studied Kaggle "Planet: Understanding the Amazon from Space" competition, whose top solutions used ensembles of CNNs (e.g., ResNet/DenseNet variants) and achieved macro F2-scores above 0.93. More broadly, remote-sensing deep learning research (e.g., studies applying CNNs to Sentinel-2 imagery for land-cover classification) has shown that combining spectral indices like NDVI with CNN features improves detection of vegetation change compared to using raw RGB imagery alone. Our project will directly compare a raw-RGB CNN against an NDVI-augmented CNN to quantify this effect on our own imbalanced label set.

## Motivation

The Amazon rainforest loses tens of thousands of square kilometers of forest cover every year, and timely detection of illegal logging, mining, and land clearing is essential for environmental enforcement, carbon-accounting efforts, and climate policy. Manually reviewing satellite imagery at the scale required

to monitor an area the size of the Amazon basin is infeasible for human analysts; an automated multi-label tagging system could flag areas of concern far faster and more consistently, supporting conservation organizations and policymakers with actionable, near-real-time information.

From a learning perspective, this project pushes us beyond single-label image classification into multi-label prediction, multi-band (non-RGB) image processing, engineered domain-specific features (NDVI), and working with the kind of severe class imbalance that is common in real-world environmental and remote-sensing datasets. It is also a chance to connect a technical machine learning project to a concrete, high-stakes societal problem.

## Evaluation

**Success criteria:** We will consider the project successful if our best model achieves a macro-averaged F2-score of at least 0.85 across all 17 labels, and, more importantly, achieves a recall of at least 0.60 on the rare but high-impact classes (mining, selective logging, slash-and-burn agriculture) despite their scarcity in the training data. We also want to demonstrate a clear, measurable improvement from adding the engineered NDVI channel compared to an RGB-only baseline.

**Technical measurement:** We will use the F2-score (which weights recall more heavily than precision, matching the intuition that missing real deforestation is more costly than a false alarm) as our primary metric, both per-label and macro-averaged, alongside per-label precision/recall curves, Hamming loss, and confusion matrices. We will also perform qualitative error analysis using Grad-CAM heatmaps on misclassified tiles to understand whether errors stem from cloud cover, mixed land use within a single tile, or genuinely ambiguous visual patterns.

## Resources

**Dataset:** "Planet: Understanding the Amazon from Space" (Kaggle), which provides roughly 40,000 labeled 256×256 pixel satellite image chips (4-band, ~20 GB total) covering the Amazon basin, each tagged with one or more of 17 possible atmospheric and land-cover/land-use labels.

**Computational tools:** Google Colab (GPU runtime) for model training; Python with TensorFlow/Keras or PyTorch for building the CNN; rasterio and Pillow for reading multi-band imagery; scikit-learn for evaluation metrics; and matplotlib/seaborn for visualizing label distributions, training curves, and confusion matrices.

**Figure 1. Deforestation Detection Pipeline**

Figure 1. Planned end-to-end pipeline: multi-band satellite chips are preprocessed and augmented with an NDVI feature, passed through a CNN multi-label classifier, and evaluated with F2, precision, and recall.

## Contributions

**Team Members:** OluwaKoretimi Adebowale | Josselin Jimenez | Mehrab U Azan

- **OluwaKoretimi Adebowale — Data & Preprocessing:** Acquire and organize the Planet Amazon dataset, normalize the 4-band imagery, engineer the NDVI feature, analyze per-label frequency to quantify class imbalance, and implement the augmentation strategy (rotation/flip, oversampling of rare classes). *Writing:* draft the Introduction and Resources sections.
- **Josselin Jimenez — Modeling:** Build and train the CNN multi-label classifier (both the from-scratch and transfer-learning/pretrained-backbone versions), tune hyperparameters (learning rate, per-label decision thresholds, class weights), and manage experiment tracking across model variants. *Writing:* draft the Motivation section and the technical "planned approach" portion of the Introduction.
- **Mehrab U Azan — Evaluation & Interpretability:** Compute per-label and macro-averaged F2-score, precision/recall, and Hamming loss; generate confusion matrices; and run the Grad-CAM error analysis comparing the RGB-only baseline against the NDVI-augmented model. *Writing:* draft the Evaluation section.

All three of us will collaborate jointly on interpreting the final results, compiling the written report, and reviewing/editing every section (including this Contributions section and the Works Cited) so the workload stays balanced and everyone has visibility into the full project, not just their own piece.

## Works Cited

Kaggle. "Planet: Understanding the Amazon from Space." kaggle.com/competitions/planet-understanding-the-amazon-from-space. Accessed 2026.

Selvaraj, A., et al. "Deep Learning Approaches for Land Cover and Land Use Classification Using Remote Sensing Imagery." Remote Sensing Applications, various issues, 2019–2023.